

**27.7** An electron has a de Broglie wavelength of  $10^{-10}$  m. Find its kinetic energy in electron volt?

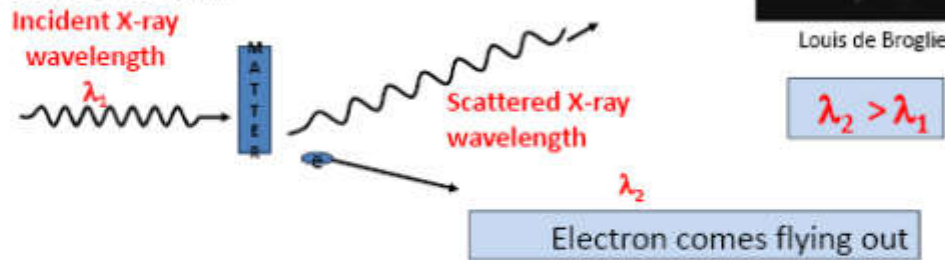
**27.8** Find the de Broglie wavelength of a baseball of mass 0.15kg moving at 30 m/s.

**27.12** An oxygen atom has a mass of 16 u, where  $1 \text{ u} = 1.66 \times 10^{-26} \text{ kg}$ . If the atom has a kinetic energy of 1 eV, what is its de Broglie wavelength?

## 26.3 X-RAYS

### The Compton Effect

In 1924, A. H. Compton performed an experiment where X-rays impinged on matter, and he measured the scattered radiation.



Louis de Broglie

Compton directed a beam of x-rays toward a block of graphite. He found that the scattered x-rays had a slightly longer wavelength than the incident x-rays. This means they also had less energy; the amount of energy reduction depended on the angle at which the x-rays were scattered. The change in wavelength is called the *Compton shift*. The Compton shift depends on the **scattering angle** and **not on the wavelength**.

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Compton effect was first observed by Arthur Compton in 1923 and this discovery led to his award of the 1927 Nobel Prize in Physics. The discovery is important because it demonstrates that light cannot be explained purely as a wave phenomenon. Compton's work convinced the scientific community that light can behave as a stream of particles (photons) whose energy is proportional to the frequency.

The change in wavelength of the scattered photon is given by:

$$hf' = hf - E_{el}$$

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)$$